



CANDIDATE
NAME

--

CENTRE
NUMBER

--	--	--	--	--

CANDIDATE
NUMBER

--	--	--	--

9231/32

October/November 2021

1 hour 30 minutes

You will need: List of formulae (MF19)

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

DC (CE/FC) 199210/2
© UCLES 2021

[Turn over

- 1 A particle is projected with speed u at an angle α above the horizontal from a point O on a horizontal plane. The particle moves freely under gravity.

- (a) Write down the horizontal and vertical components of the velocity of the particle at time T after projection. [2]

$$V_y = u \sin \alpha - gT$$

$$V_x = u \cos \alpha$$

At time T after projection, the direction of motion of the particle is perpendicular to the direction of projection.

- (b) Express T in terms of u , g and α . [2]

$$y = x \tan \alpha - \frac{gx^2}{2u^2} \sec^2 \alpha$$

$$\frac{dy}{dx} = \tan \alpha - \frac{gx}{u^2} \sec^2 \alpha$$

$$\text{@ time } T, x = uT \cos \alpha$$

$$\rightarrow \frac{dy}{dx} = \tan \alpha - \frac{g u T \cos \alpha}{u^2 \cos^2 \alpha}$$

$$= \tan \alpha - \frac{gT}{u \cos \alpha}$$

$$\text{Initially, } \frac{dy}{dx} = \tan \alpha$$

$$\rightarrow \text{so at } T, \frac{dy}{dx} = \frac{-1}{\tan \alpha}$$

$$\frac{-1}{\tan \alpha} = \tan \alpha - \frac{gT}{u \cos \alpha}$$

$$\frac{gT}{u \cos \alpha} = \frac{\tan^2 \alpha + 1}{\tan \alpha}$$

$$T = \left(\frac{1}{\cos^2 \alpha} \right) \left(\frac{\cos \alpha}{\sin \alpha} \right) \left(\frac{u \cos \alpha}{g} \right)$$

$$= \frac{u}{g \sin \alpha}$$

- (c) Deduce that $T > \frac{u}{g}$. [1]

$$\frac{u}{g \sin \alpha} > \frac{u}{g}$$

$$1 > \sin \alpha$$

$$\rightarrow \alpha \neq 90^\circ, \text{ so this is true}$$

- 2 A light spring AB has natural length a and modulus of elasticity $5mg$. The end A of the spring is attached to a fixed point on a smooth horizontal surface. A particle P of mass m is attached to the end B of the spring. The spring and particle P are at rest on the surface.

Another particle Q of mass km is moving with speed $\sqrt{4ga}$ along the horizontal surface towards P in the direction BA . The particles P and Q collide directly and coalesce. In the subsequent motion the greatest amount by which the spring is compressed is $\frac{1}{5}a$.

Find the value of k .

[6]

Diagram: A horizontal line represents a spring of natural length a and modulus of elasticity $5mg$. A particle P of mass m is attached to the end B . A particle Q of mass km is moving towards P with speed $\sqrt{4ga}$. The particles collide and coalesce. The coefficient of restitution $e = 0$.

Method: Conservation of energy

Initially:

$$KE = \frac{1}{2}(km+m)(v)^2 = \frac{1}{2}(k+1)m\left(\frac{4ga k^2}{(k+1)^2}\right) = \frac{1}{2}m\left(\frac{4gak^2}{(k+1)}\right)$$

Finally:

$$KE = \frac{1}{2}(km+m)(v)^2$$

$$EPE = \frac{\lambda x^2}{2\lambda} = \frac{(5mg)(\frac{a}{5})^2}{2a} = \frac{5mga^2}{50a} = \frac{1}{10}mga$$

But: At the point of maximum compression

↳ particle is turning

↳ $\therefore v=0, KE=0$

So, by CoE:

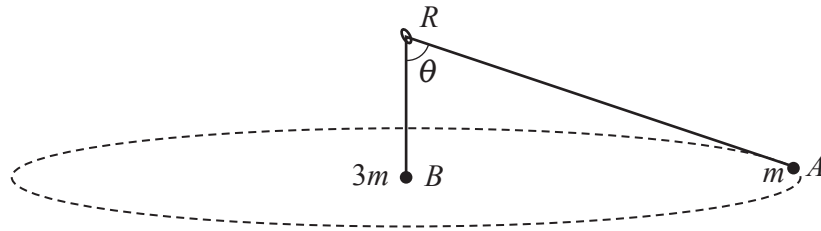
$$\frac{mga}{10} = \frac{1}{2}m\left(\frac{4gak^2}{(k+1)}\right)$$

$$\frac{1}{20} = \frac{k^2}{k+1} \Rightarrow 20k^2 - k - 1 = 0$$

$$(4k-1)(5k+1) = 0$$

↳ $k = \frac{1}{4}$ (as $k > 0$)

3

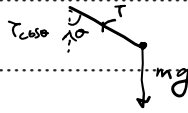


Particles A and B , of masses m and $3m$ respectively, are connected by a light inextensible string of length a that passes through a fixed smooth ring R . Particle B hangs in equilibrium vertically below the ring. Particle A moves in horizontal circles with speed v . Particles A and B are at the same horizontal level. The angle between AR and BR is θ (see diagram).

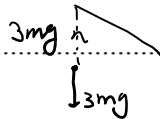
(a) Show that $\cos \theta = \frac{1}{3}$.

[2]

$$T \cos \theta = mg$$



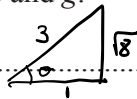
$$T = 3mg$$



$$\cos \theta = \frac{1}{3}$$

(b) Find an expression for v in terms of a and g .

[4]



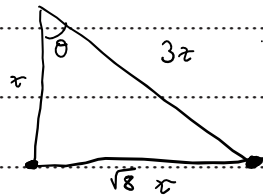
$$\frac{mv^2}{r} = T \sin \theta$$

$$\frac{mv^2}{r} = 3mg \left(\frac{2\sqrt{2}}{3} \right)$$

$$v^2 = 2\sqrt{2} rg$$

$$v^2 = 2\sqrt{2} \left(\frac{a}{3} \right) g$$

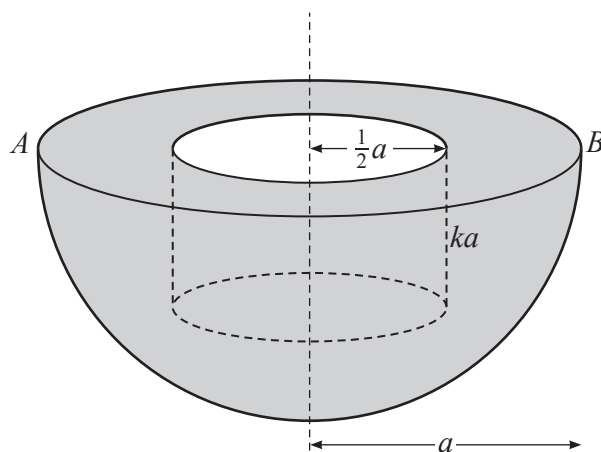
$$v = \sqrt{2ga}$$



$$x + 3x = a$$

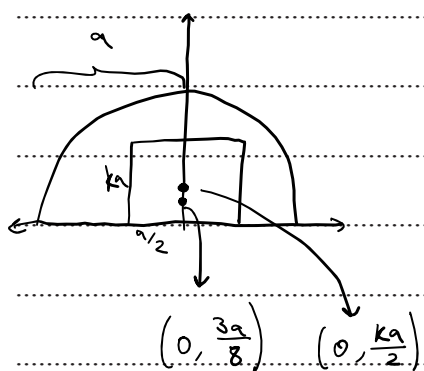
$$\hookrightarrow x = \frac{a}{4}$$

$$r = \sqrt{8}x = \frac{\sqrt{2}a}{4} = \frac{a}{2\sqrt{2}}$$



An object is formed by removing a solid cylinder, of height ka and radius $\frac{1}{2}a$, from a uniform solid hemisphere of radius a . The axes of symmetry of the hemisphere and the cylinder coincide and one circular face of the cylinder coincides with the plane face of the hemisphere. AB is a diameter of the circular face of the hemisphere (see diagram).

- (a) Show that the distance of the centre of mass of the object from AB is $\frac{3a(2-k^2)}{2(8-3k)}$. [4]



Taking moments:

$$\bar{y} \left(\frac{2}{3} \pi a^3 - \pi \left(\frac{a}{2} \right)^2 (ka) \right) (p) \\ = \left(\frac{2}{3} \pi a^3 \right) (p) \left(\frac{3a}{8} \right) + \left(-\pi \left(\frac{a}{2} \right)^2 (ka) \right) (p) \left(\frac{ka}{2} \right)$$

$$\bar{y} \left(\pi a^3 p \right) \left(\frac{2}{3} - \frac{k}{4} \right) \\ = \pi a^3 p \left(\frac{a}{4} \right) - \pi a^3 p \left(\frac{k^2 a}{8} \right)$$

$$\bar{y} \left(\frac{8-3k}{3} \right) = \frac{2a-k^2 a}{2}$$

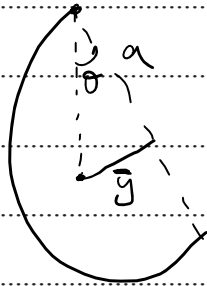
$$\bar{y} = \frac{6a - 3k^2 a}{16 - 6k}$$

$$= \frac{3a(2-k^2)}{2(8-3k)}$$

When the object is freely suspended from the point A , the line AB makes an angle θ with the downward vertical, where $\tan \theta = \frac{7}{18}$.

(b) Find the possible values of k .

[3]



$$\tan \theta = \frac{7}{18}$$

$$\frac{y}{a} = \frac{7}{18}$$

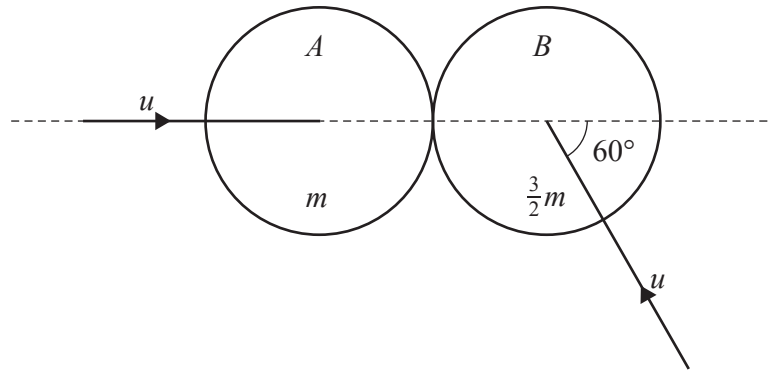
$$\left(\frac{3}{2}\right)\left(\frac{2-k^2}{8-3k}\right) = \frac{7}{18}$$

$$108 - 54k^2 = 112 - 42k$$

$$0 = 54k^2 - 42k + 4$$

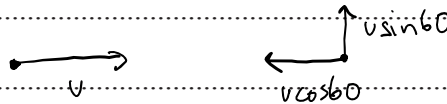
$$= (3k-2)(9k-1)(2)$$

$$\rightarrow k = \frac{2}{3} \text{ or } \frac{1}{9}$$



Two uniform smooth spheres A and B of equal radii have masses m and $\frac{3}{2}m$ respectively. The two spheres are each moving with speed u on a horizontal surface when they collide. Immediately before the collision A 's direction of motion is along the line of centres, and B 's direction of motion makes an angle of 60° with the line of centres (see diagram). The coefficient of restitution between the spheres is $\frac{2}{3}$.

- (a) Find the angle through which the direction of motion of B is deflected by the collision. [6]



Only horizontal components are affected by the collisions

$$\text{CoM: } mu - \left(\frac{3m}{2}\right)\left(\frac{u}{2}\right) = mv_1 + \frac{3m}{2}v_2$$

$$\frac{u}{4} = v_1 + \frac{3}{2}v_2$$

$$\text{Newtons: } \frac{v_2 - v_1}{\frac{3u}{2}} = \frac{2}{3}$$

$$v_2 - v_1 = u$$

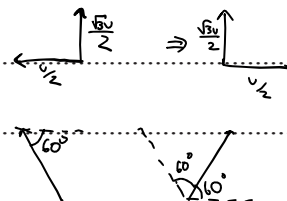
$$+ v_1 + \frac{3}{2}v_2 = \frac{u}{4}$$

$$\frac{5}{2}v_2 = \frac{5u}{4}$$

$$\boxed{v_2 = \frac{u}{2}}$$

$$v_1 = -\frac{u}{2}$$

Considering B post collision



It is deflected through

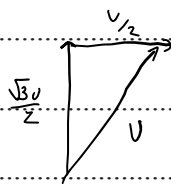
a $\boxed{60^\circ}$ angle

- (b) Find the loss in the total kinetic energy of the system as a result of the collision. [3]

Initially:

$$KE = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{3m}{2}\right)(v^2) = \frac{5}{4}mv^2 \text{ J}$$

Find:



$$\rightarrow B \text{ has } \frac{1}{2}\left(\frac{3m}{2}\right)v^2 = \frac{3m}{4}v^2 \text{ J}$$

$$\leftarrow \frac{v}{2} \rightarrow A \text{ has } \frac{1}{2}(m)\left(\frac{v}{2}\right)^2 = \frac{mv^2}{8} \text{ J}$$

$$Loss = \frac{5mv^2}{4} - \frac{3mv^2}{4} - \frac{mv^2}{8} = \boxed{\frac{3mv^2}{8}} \text{ J}$$

- 6 A particle P of mass 2 kg moves along a horizontal straight line. The point O is a fixed point on this line. At time $t\text{ s}$ the velocity of P is $v\text{ ms}^{-1}$ and the displacement of P from O is $x\text{ m}$.

A force of magnitude $\left(8x - \frac{128}{x^3}\right)\text{ N}$ acts on P in the direction OP . When $t = 0$, $x = 8$ and $v = -15$.

- (a) Show that $v = -\frac{2}{x}(x^2 - 4)$. [5]

$$a = \left(8x - \frac{128}{x^3}\right)\left(\frac{1}{x}\right) = 4x - \frac{64}{x^3}$$

By Chain rule:

$$\frac{\left(\frac{dv}{dt}\right)}{\left(\frac{dx}{dt}\right)} = \frac{4x - \frac{64}{x^3}}{dx}$$

$$v dv = 4x - \frac{64}{x^3} dx$$

$$\int v dv = \int 4x - \frac{64}{x^3} dx$$

$$\frac{v^2}{2} = 2x^2 + \frac{32}{x^2} + C$$

$$v^2 = 4x^2 + \frac{64}{x^2} + C$$

$$\text{sub } (8, -15)$$

$$225 = 4(64) + 1 + C$$

$$C = -32$$

$$v^2 = 4x^2 + \frac{64}{x^2} - 32$$

$$= (2x)^2 + \left(\frac{8}{x}\right)^2 - 2(2x)\left(\frac{8}{x}\right)$$

$$v^2 = \left(2x - \frac{8}{x}\right)^2$$

$$v^2 = \left(\frac{2}{x}(x^2 - 4)\right)^2$$

$$v = \pm \left(\frac{2}{x}(x^2 - 4)\right)$$

$\Rightarrow$ we know that when $x = 8$, $v = -15$

$\hookrightarrow$ we take the negative root

$$v = -\frac{2}{x}(x^2 - 4)$$

(b) Find an expression for x in terms of t .

[4]

$$\frac{dx}{dt} = \frac{-2}{x} (x^2 - 4)$$

$$\frac{-x}{2} \left(\frac{1}{x^2 - 4} \right) dx = dt$$

$$\int \frac{x}{x^2 - 4} dx = \int -2 dt$$

$$\int \frac{x}{x^2 - 4} dx = -2t + C$$

$$\begin{aligned} \hookrightarrow \text{let } u &= x^2 - 4 \\ du &= 2x dx \\ \hookrightarrow dx &= \frac{du}{2x} \end{aligned}$$

$$\begin{aligned} \int \frac{x}{x^2 - 4} dx &= \int \frac{x}{u} \left(\frac{du}{2x} \right) \\ &= \frac{1}{2} \int \frac{1}{u} du \\ &= \frac{1}{2} \ln|u| + C \end{aligned}$$

$$= \frac{1}{2} \ln|x^2 - 4| + C$$

$$\therefore \frac{1}{2} \ln|x^2 - 4| = -2t + C$$

$$\textcircled{a} t = 0, x = 8$$

$$\frac{1}{2} \ln|60| = C$$

$$\frac{1}{2} \ln|x^2 - 4| = -2t + \frac{1}{2} \ln 60$$

$$\Rightarrow 64 - 4 = 60e^{-4t}, \text{ not } -60e^{-4t}$$

$$\ln|x^2 - 4| = -4t + \ln 60$$

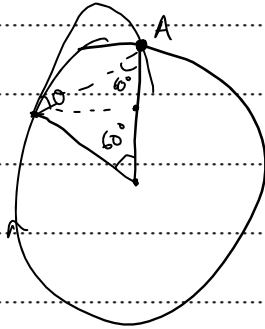
$$|x^2 - 4| = e^{-4t + \ln 60}$$

$$|x^2 - 4| = 60e^{-4t}$$

$$\therefore x^2 - 4 = 60e^{-4t} \Rightarrow x = \sqrt{4 + 60e^{-4t}} \quad \text{as}$$

- 7 One end of a light inextensible string of length a is attached to a fixed point O . The other end of the string is attached to a particle P of mass m . The particle P is held vertically below O with the string taut and then projected horizontally. When the string makes an angle of 60° with the upward vertical, P becomes detached from the string. In its subsequent motion, P passes through the point A which is a distance a vertically above O .

- (a) The speed of P when it becomes detached from the string is V . Use the equation of the trajectory of a projectile to find V in terms of a and g . [4]



$$y = x \tan \theta - \frac{gx^2}{2v^2} \sec^2 \theta$$

$$\text{initially: } x = y = 0 \Rightarrow (0, 0)$$

$$\text{At A: } x = a \sin 60 = \frac{\sqrt{3}}{2}a \Rightarrow \left(\frac{\sqrt{3}a}{2}, \frac{a}{2} \right)$$

$$y = a - a \cos 60 = \frac{a}{2}$$

a point

$$\rightarrow \text{Also } \theta = 60^\circ$$

$$\rightarrow \text{Substituting } \left(\frac{\sqrt{3}a}{2}, \frac{a}{2} \right) \text{ and } \theta = 60^\circ:$$

$$\frac{a}{2} = \frac{\sqrt{3}}{2}a \tan 60 - \frac{g \left(\frac{\sqrt{3}a}{2} \right)^2}{2(v^2)} \sec^2(60)$$

$$a = \sqrt{3} \left(\frac{\sqrt{3}}{2}a \right) - \frac{g \left(\frac{3a^2}{4} \right)}{v^2} \quad (4)$$

$$a = 3a - \frac{3a^2g}{v^2}$$

$$\frac{3a^2g}{v^2} = 2a$$

$$2v^2 = 3ga$$

$$v = \sqrt{\frac{3}{2}ga}$$

- (b) Find, in terms of m and g , the tension in the string immediately after P is initially projected horizontally. [4]

$$\frac{mv^2}{r} = \frac{mv^2}{r} - 2mg + 2mg \cos \theta$$

$$\hookrightarrow @ \theta = 120^\circ, v = \sqrt{\frac{3}{2}ga}$$

$$\frac{m(\frac{3}{2}ga)}{a} = \frac{mv^2}{a} - 2mg + 2mg \cos 120$$

$$\frac{3}{2}g = \frac{v^2}{a} - 3g$$

$$v = \sqrt{4.5ga}$$

$$T = \frac{mv^2}{r} - 2mg + 3mg \cos \theta$$

$$\hookrightarrow \theta = 0, v = \sqrt{4.5ga}$$

$$T = \frac{m(4.5ga)}{a} - 2mg + 3mg$$

$$= 4.5mg - 2mg + 3mg$$

$$\boxed{\approx 5.5mg}$$

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines, providing a guide for letter height and placement. The lines are evenly spaced across the entire page, leaving ample room for writing practice. There is no text or other markings on the page.

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.